

Technology Stack

Date	30 June 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

The **Smart Lender** project uses **HTML**, **CSS**, and **Jinja2** for the frontend, **Python Flask** for the backend, **Scikit-learn** for machine learning, **NumPy** for data processing, and **Pickle** to load the trained model. **GitHub** is used for version control and collaboration. This technology stack ensures a simple, efficient, and reliable loan prediction system.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 Templates	Responsive user interface for loan prediction and dynamic rendering.

2	Backend / Server-Side	Python, Flask	Lightweight framework for handling routes, prediction logic, and web requests.
3	Database / Data Storage	Scikit-learn, Pickle	Loads the trained loan prediction model and performs predictions efficiently.
4	Cloud / Hosting / Deployment	NumPy	Converts input into numerical arrays and preprocesses data before prediction.
5	Version Control & CI/CD	StandardScaler (scikit-learn)	Normalizes user input before feeding it to the trained model.

6	Third-Party APIs / Other Tools	VS Code, Flask Development Server	Development, debugging, and application testing.
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